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BLE Expert Protocol Sniffer

A Wireshark-like Tool for Bluetooth Low Energy Analysis

Technical Stack

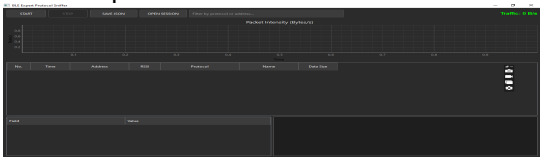
Built using Python 3.12 with **bleak** for BLE scanning, **PyQt6** for the GUI, and **pyqtgraph** for real-time visualization. Uses multi-threading with Qt signals for thread-safe communication.

Implementation

Protocol identification works by mapping manufacturer IDs to device types: Apple (0x004C), Microsoft (0x0006), Samsung (0x0075), Google (0x00E0). Packets display timestamp, MAC address, RSSI, protocol, device name, and data size.

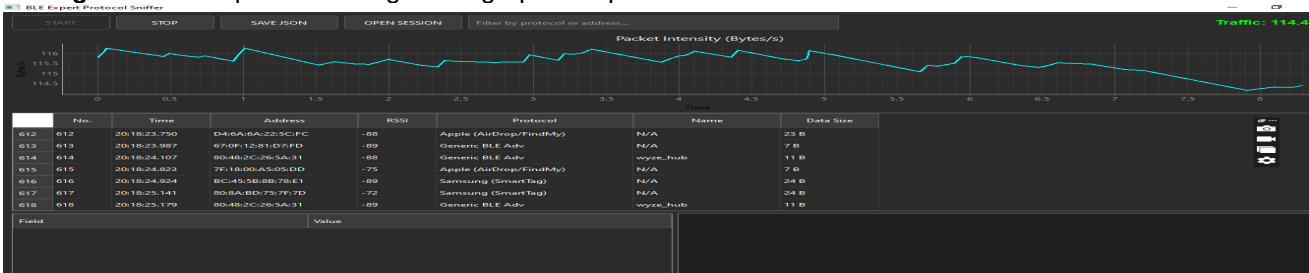
Application Screenshots

Fig 1: Application startup with control panel



Dark-themed interface with START/STOP controls, save/load buttons, and filter input. Traffic at 0 B/s.

Fig 2: Active capture showing traffic graph and protocol detections



Packet Intensity graph shows 114.4 B/s traffic. Captured 600+ packets from Apple, Samsung, and Generic BLE devices. Wyze_hub identified.

Fig 3: Deep packet inspection with tree view and manufacturer data

Field	Value	Raw Data
BLE Advertisement		
Source	5D:79:A2:90:50:CC	
TX Power	0 dBm	
Manufacturer Specific Data		
Company ID: 0x004C	01000000000000000000000000000000	01000000000000000000000000000000

Tree view shows BLE Advertisement structure: Source address, TX Power (0 dBm), and Manufacturer Data. Company ID 0x004C identifies Apple. Raw hex displayed on right.

Fig 4: Apple AirDrop/FindMy protocol packets from iOS devices

No.	Time	Address	RSSI	Protocol	Name	Data Size
3109	20:26:52.710	80:48:2C:26:5A:31	-95	Generic BLE Adv	wyze_hub	11 B
3110	20:26:52.710	7F:18:00:A5:05:DD	-66	Apple (AirDrop/FindMy)	N/A	23 B
3111	20:26:52.711	7F:18:00:A5:05:DD	-66	Apple (AirDrop/FindMy)	N/A	23 B
3112	20:26:52.812	5F:93:7D:0D:17:5E	-27	Apple (AirDrop/FindMy)	N/A	7 B
3113	20:26:52.836	5F:93:7D:0D:17:5E	-27	Apple (AirDrop/FindMy)	N/A	7 B
3114	20:26:52.851	7F:18:00:A5:05:DD	-72	Apple (AirDrop/FindMy)	N/A	23 B
3115	20:26:52.855	7F:18:00:A5:05:DD	-71	Apple (AirDrop/FindMy)	N/A	23 B

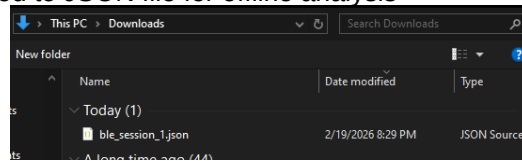
Apple devices broadcasting 7B and 23B advertisements for AirDrop and FindMy. RSSI from -66 to -95 dBm.

Fig 5: Samsung SmartTag packets with 24-byte payloads

No.	Time	Address	RSSI	Protocol	Name	Data Size
3984	20:28:08.042	80:8A:BD:75:7F:7D	-82	Samsung (SmartTag)	N/A	24 B
3985	20:28:08.156	80:48:2C:26:5A:31	-88	Generic BLE Adv	wyze_hub	11 B
3986	20:28:08.425	1C:86:9A:61:98:7B	-90	Samsung (SmartTag)	N/A	24 B
3987	20:28:08.538	BC:45:5B:8B:78:E1	-80	Samsung (SmartTag)	N/A	24 B
3988	20:28:08.568	5F:93:7D:0D:17:5E	-28	Apple (AirDrop/FindMy)	N/A	7 B
3989	20:28:08.584	80:48:2C:26:5A:31	-91	Generic BLE Adv	wyze_hub	11 B
3990	20:28:08.661	80:48:2C:26:5A:31	-90	Generic BLE Adv	wyze_hub	11 B

Samsung SmartTag devices broadcasting 24B packets. RSSI ranged from -82 to -28 dBm. Wyze_hub device detected with 11B payloads at -88 to -91 dBm.

Fig 6: Captured session saved to JSON file for offline analysis



Session saved to ble_session_1.json with all packet metadata for offline analysis.

Challenges and Solutions

- 1. Asynchronous BLE Scanning:** Used QThread for background scanning with Qt signals for thread-safe updates.
- 2. Protocol Identification:** Mapped company IDs to protocol names using Bluetooth SIG assignments.
- 3. JSON Serialization:** Converted bytes to hex strings before saving to preserve data.
- 4. Real-time Visualization:** Implemented data throttling to maintain responsive interface.

What I Learned

Key concepts learned:

BLE Protocol Structure: Advertisement packets, manufacturer data, service UUIDs, and device broadcasting.

Asynchronous Programming: Managing async BLE operations with threading and signal-slot patterns.

GUI Development: Building responsive applications with PyQt6 tables, tree views, and graphs.

Protocol Analysis: Identifying manufacturers and protocols from raw packet data.

Data Visualization: Real-time graphing with pyqtgraph for traffic patterns.

Conclusion

The BLE sniffer successfully captures and analyzes Bluetooth traffic from multiple device types. Testing detected Apple AirDrop/FindMy, Samsung SmartTags, and generic BLE devices, capturing 3000+ packets with protocol information. Future enhancements: RSSI filtering, PCAP export, and statistical analysis.